

COM4573 • ARTIFICIAL NEURAL NETWORKS

Week 4 • Activation & Loss Functions

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Question of the week

A model predicts 0.95. Is that good?

It depends on the true label, the task, and how the output is interpreted.

Think • pair • share

What should happen if the true label is 0 instead of 1?

From MLP to learning

Last week

x → weighted sum → ReLU hidden representation → output.

This week

Choose the output activation and quantify prediction error.

Next week

Use derivatives to send learning signals backward through the network.

Activation versus loss

Activation f(z)

Transforms a neuron’s pre-activation z into an output; affects range and nonlinearity.

Loss L(y, ŷ)

Compares a prediction with its target and produces a numerical penalty.

Why both?

A useful output representation and a suitable training objective must work together.

Step and sigmoid

Step

Returns 0 or 1 at a threshold; useful for the classic perceptron, but not differentiable at the threshold.

Sigmoid σ(z) = 1/(1+e^(−z))

Maps a real-valued score into (0,1); commonly used for binary outputs.

Worked example

σ(0)=0.5; σ(2)≈0.881; σ(−2)≈0.119.

Reading an activation formula

Symbols first: what enters the neuron and what leaves it

Sigmoid $\sigma(z) = 1 / (1 + e^{(-z)})$

z = weighted sum plus bias ($w^T x + b$).

e = exponential constant, approximately 2.718.

σ(z) = transformed output between 0 and 1.

Example: z = 0 → σ(0) = 0.5.

ReLU $f(z) = \max(0, z)$

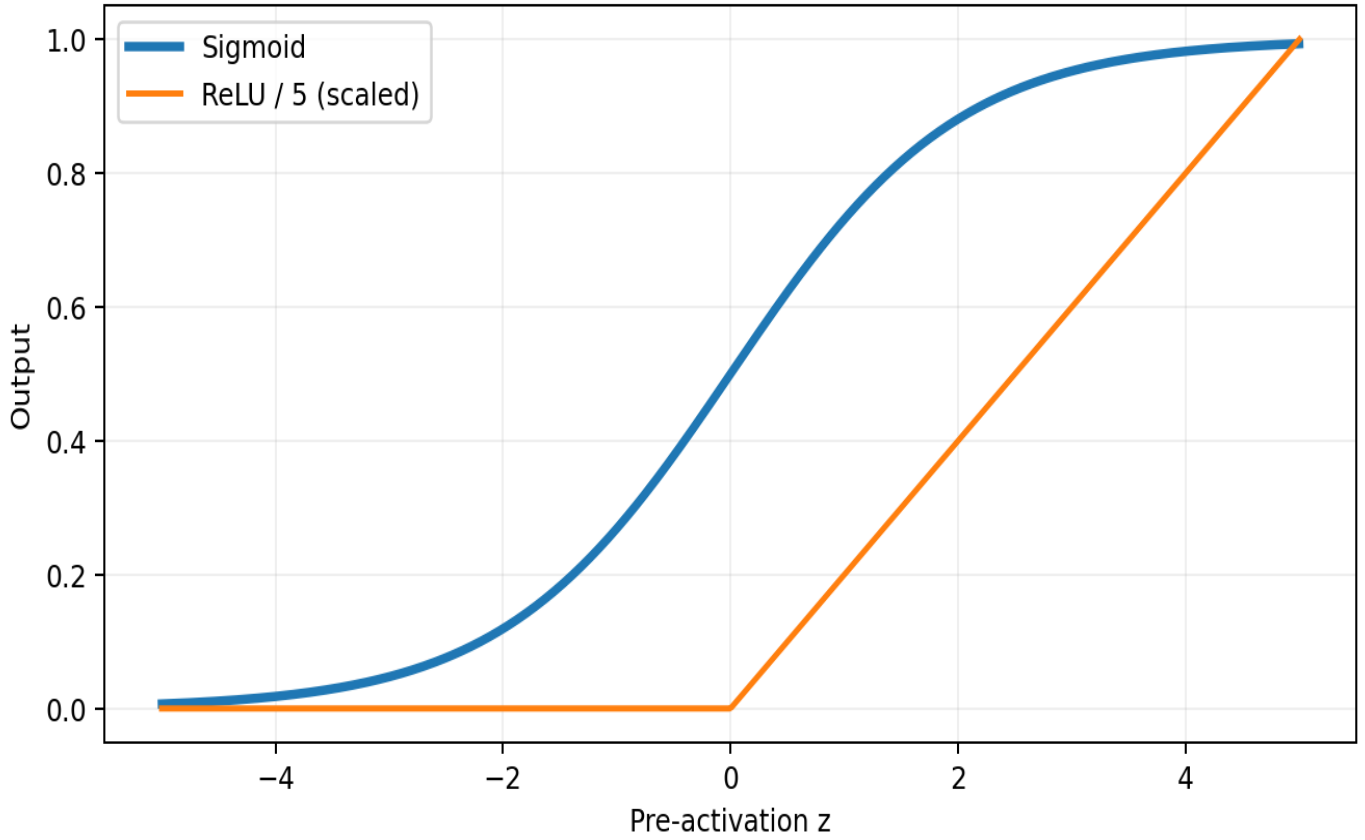
max chooses the larger of 0 and z.

If z = −2, the output is 0.

If z = 3, the output is 3.

In hidden layers, this allows nonlinear feature transformations.

Sigmoid: read the curve



A smooth probability-like output

- At z = 0, sigmoid = 0.5.
- Large positive z approaches 1.
- Large negative z approaches 0.

Tanh and ReLU

Tanh

tanh(z) ranges from −1 to 1 and is centered around zero.

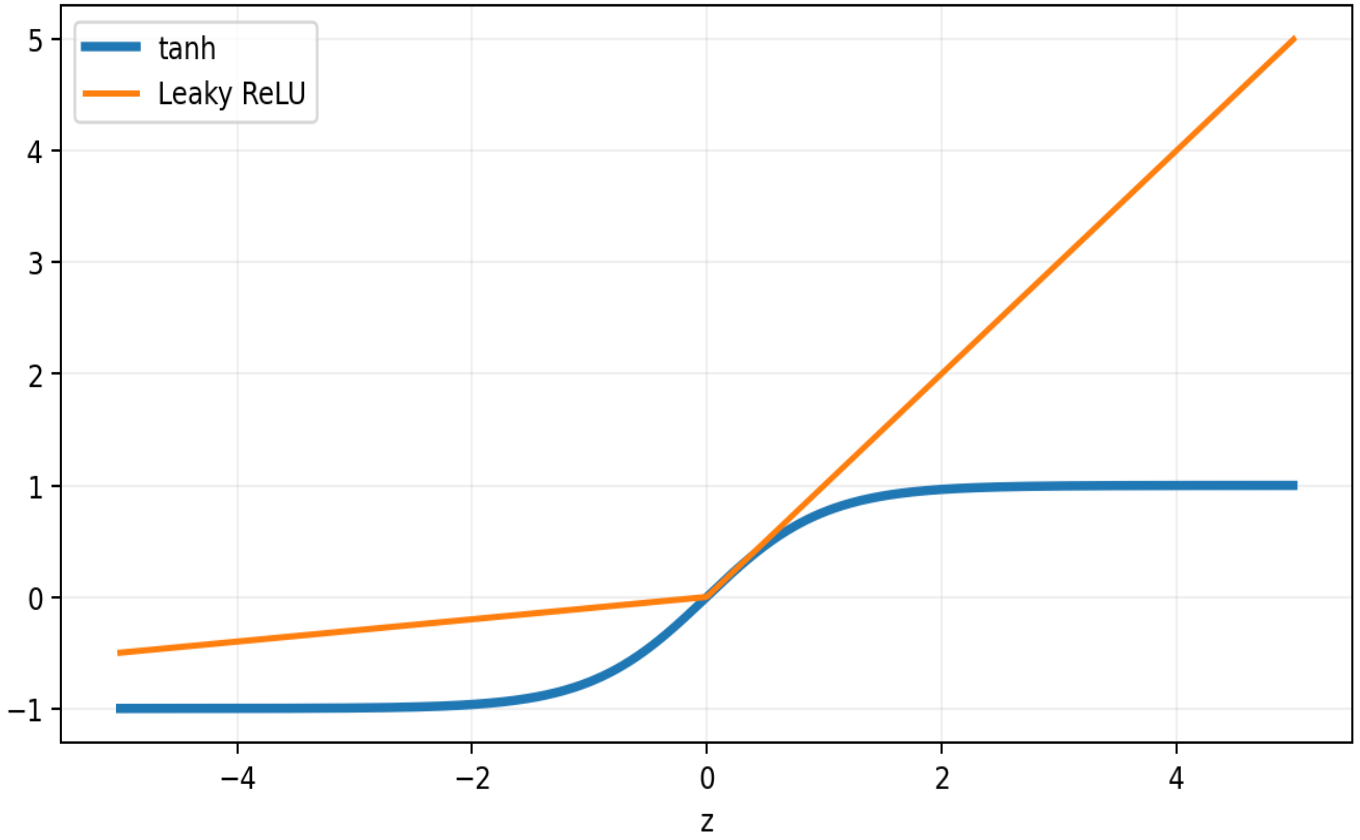
ReLU

max(0,z): negative inputs become zero; positive inputs pass through.

Check

ReLU(−3)=0, ReLU(1.5)=1.5; tanh(0)=0.

Tanh and Leaky ReLU



Two different output behaviors

- Tanh is bounded between −1 and 1.
- Leaky ReLU keeps a small negative slope.
- Activation choice affects gradients.

Leaky ReLU and GELU

Leaky ReLU

$f(z)=\max(\alpha z,z)$, often $\alpha=0.01$; keeps a small negative slope.

GELU

$f(z)=z\Phi(z)$, where Φ is the standard normal CDF; a smooth, input-dependent gate.

Modern context

GELU is used in some Transformer architectures, including BERT; modern models also use alternatives such as SiLU/SwiGLU.

Why saturation matters

Sigmoid and tanh

At very large positive or negative inputs, derivatives approach zero.

ReLU

Positive-side derivative is 1; on the negative side it is 0, so some units may stop updating.

Practical lesson

Activation selection affects gradient flow, not just output range.

Softmax is different

Multiclass output

Softmax converts a vector of class logits into positive values summing to 1.

Formula

$p_i = \exp(z_i) / \Sigma_j \exp(z_j)$; stable implementations subtract max(z).

Example

Scores [2,1,0] → probabilities ≈ [0.665, 0.245, 0.090].

Softmax formula, explained

One vector of logits becomes a probability distribution

$p_i = \exp(z_i) / \Sigma_j \exp(z_j)$

z_i: raw score (logit) for class i.

exp: exponential function.

Σ_j: add exponentials over ALL classes.

p_i: probability assigned to class i.

All p_i sum to 1.

Worked example

Logits: [2, 1, 0]

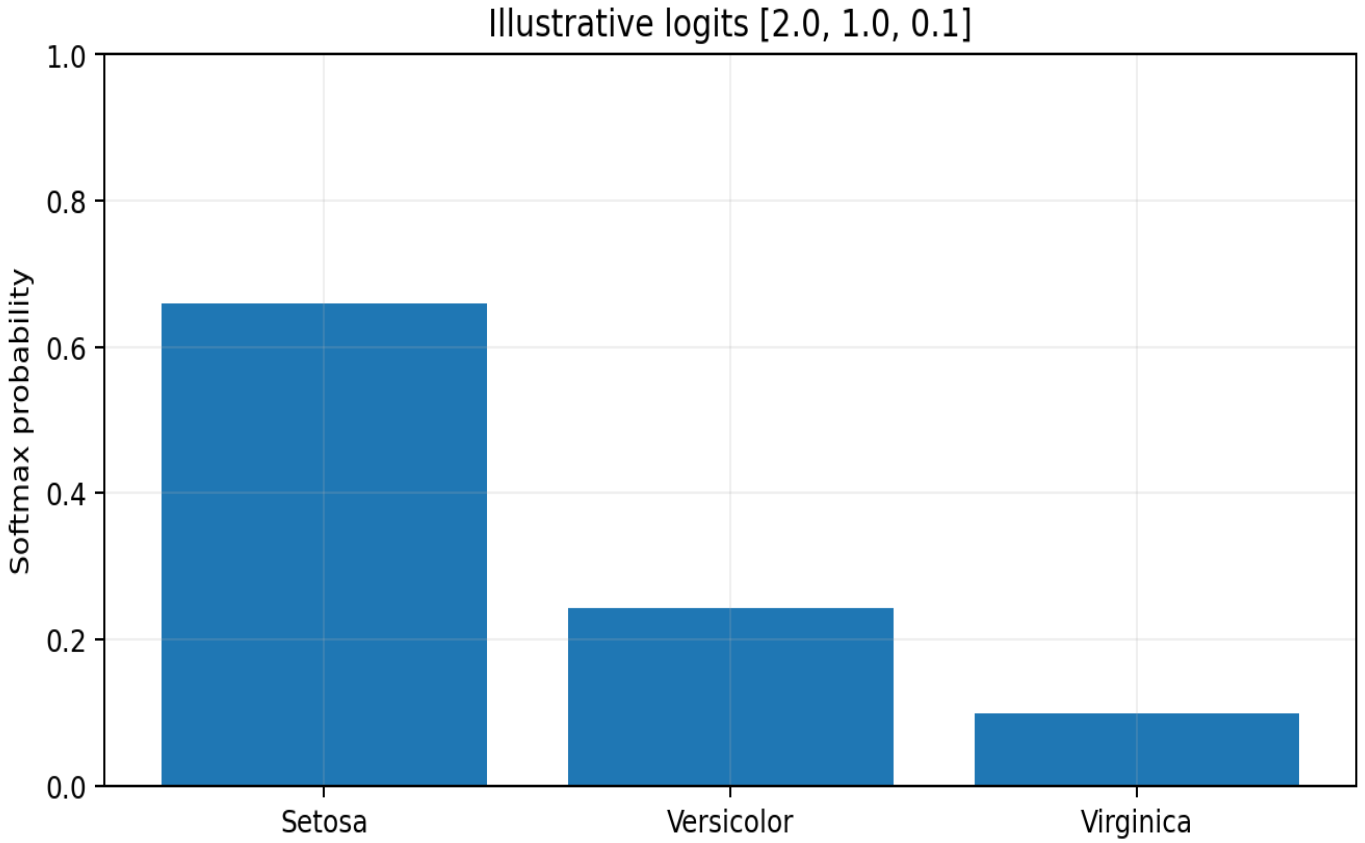
Exponentials: [7.389, 2.718, 1]

Sum: 11.107

Probabilities: [0.665, 0.245, 0.090]

The predicted class is the one with the largest probability.

From Iris scores to probabilities



Three classes, one distribution

- Softmax probabilities sum to 1.
- The highest score gives the predicted class.
- Confidence does not guarantee correctness.

Iris: from logits to classes

Network

4 Iris measurements → Dense(8, ReLU) → Dense(3, Softmax).

Suppose logits are [2,1,0]

Softmax returns ≈ [0.665, 0.245, 0.090]; class 0 has the highest score.

Important distinction

The predicted class is not the same as the loss; the true class is also needed.

Which activation should we use?

Hidden layer in a basic MLP

ReLU is a practical starting point; alternatives may work better depending on the architecture.

Binary output

One sigmoid unit is common.

Mutually exclusive multiclass output

One logit per class; softmax for interpreted probabilities.

What does a loss function measure?

Prediction and target

The same prediction may be good for one target and poor for another.

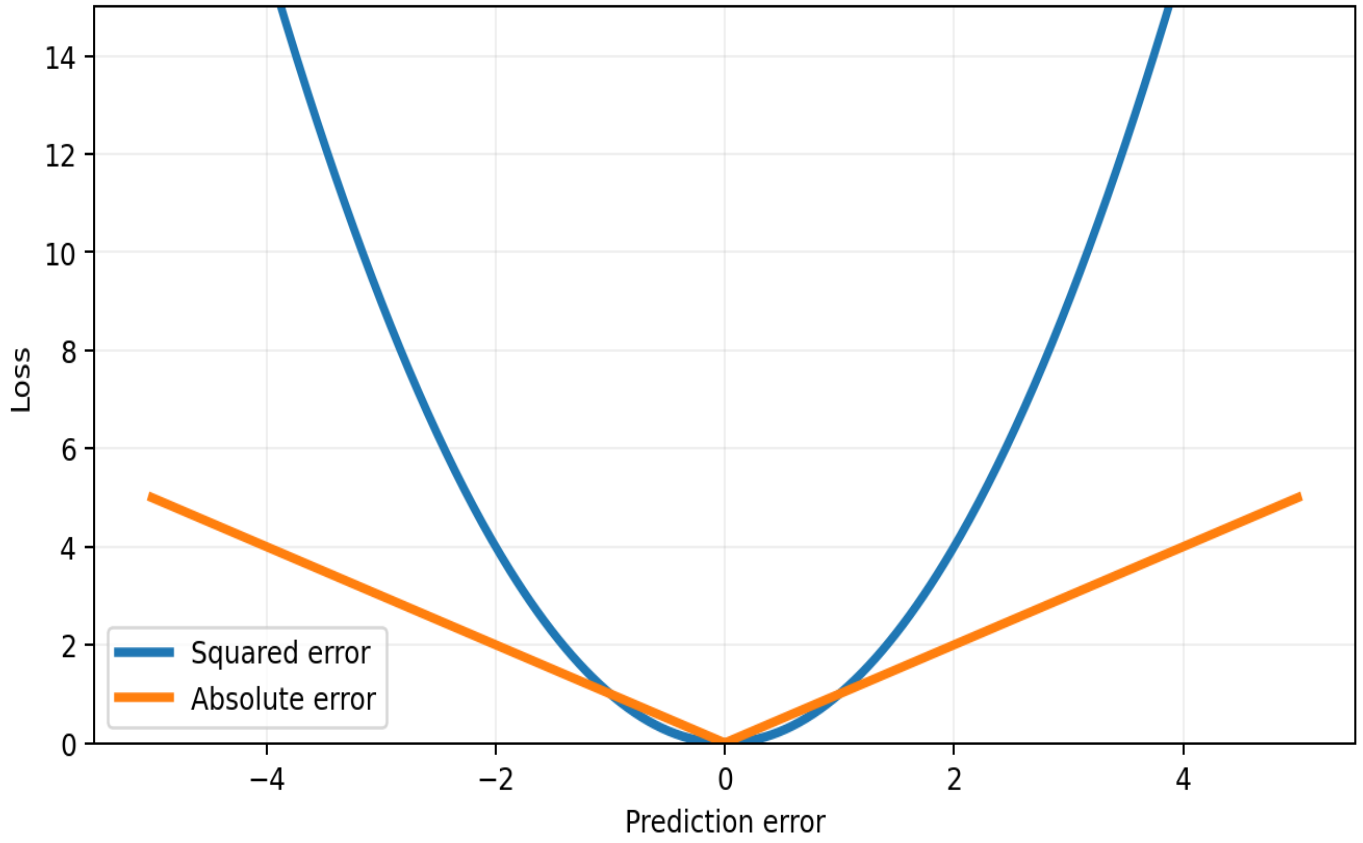
Training objective

Usually minimize the average loss over training examples.

Remember

Loss is a training objective; accuracy, F1 and other metrics answer different evaluation questions.

MSE versus MAE



What happens to a large error?

- MSE squares the error.
- MAE grows linearly with error magnitude.
- Discuss outliers in sensor readings.

Mean squared error (MSE)

Formula

MSE = $(1/n)\ \Sigma(y_i - \hat{y}_i)^2$.

Example

Targets [2,4], predictions [3,2] → squared errors [1,4] → MSE=2.5.

Interpretation

Large errors receive disproportionately larger penalties.

Understanding regression loss notation

What do n, y_i and $\hat{y}_i$ actually mean?

$MSE = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$

n: number of evaluated observations.

y_i: true numerical target for observation i.

$\hat{y}_i$: predicted target for observation i.

Σ: add errors across observations.

Squaring makes large errors especially costly.

$MAE = (1/n)\ \Sigma\ |y_i - \hat{y}_i|$

Vertical bars | | mean absolute value.

Example: y = 4, prediction = 2 → |4−2| = 2.

Both losses average over n observations.

MSE has squared target units; MAE retains original target units.

Mean absolute error (MAE)

Formula

MAE = $(1/n)\ \Sigma|y_i - \hat{y}_i|$.

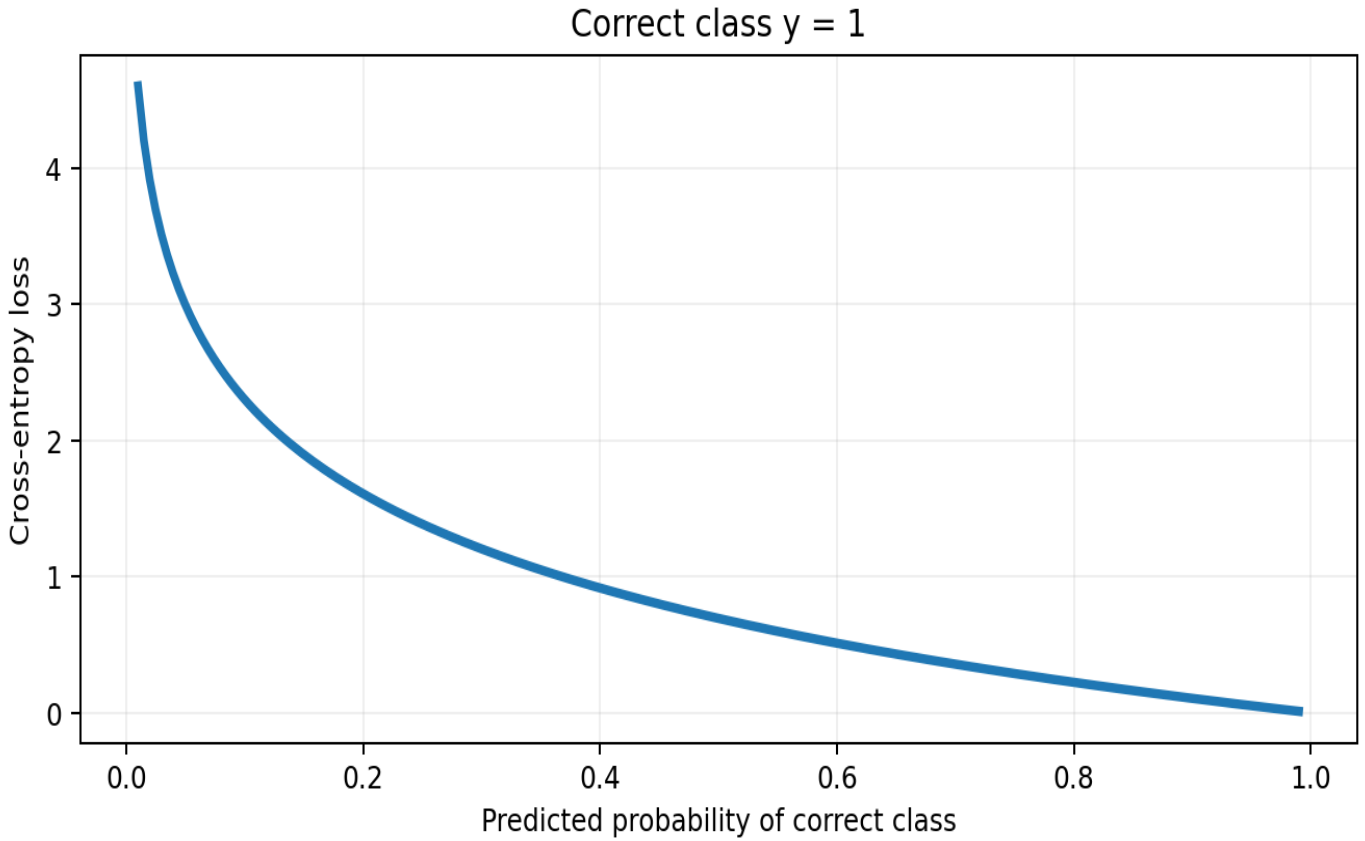
Same example

Absolute errors [1,2] → MAE=1.5.

Trade-off

MAE is less sensitive to large errors than MSE; it is not differentiable exactly at zero error.

Why confident mistakes are costly



Cross-entropy for y = 1

- p = 0.9 gives loss ≈ 0.105.
- p = 0.1 gives loss ≈ 2.303.
- Loss is not classification accuracy.

MSE versus MAE: a fair comparison

Same three errors, different penalties

Errors: [1, 1, 6]

MAE = (1 + 1 + 6)/3 = 2.67

MSE = $(1^2 + 1^2 + 6^2)/3$ = 12.67

One large error dominates MSE.

The numerical values have different units, so do not compare 2.67 vs 12.67 as if they were the same scale.

What should we choose?

MSE: emphasizes large deviations; smooth gradients; sensitive to outliers.

MAE: more robust to extreme errors; less smooth at zero.

For noisy sensor measurements, compare both on validation data and consider the cost of large misses.

Binary cross-entropy (BCE)

Formula

$L = -[y \ln(p) + (1-y) \ln(1-p)]$.

Correct confident prediction

$y=1$, $p=0.9 \rightarrow L \approx 0.105$.

Confident mistake

$y=1$, $p=0.1 \rightarrow L \approx 2.303$; wrong confident predictions cost much more.

Binary cross-entropy: decode the symbols

For a single binary-labelled observation

$L = -[y \ln(p) + (1-y) \ln(1-p)]$

$y \in \{0,1\}$: true class.

p: predicted probability for class 1.

ln: natural logarithm.

If $y = 1$, loss becomes $-\ln(p)$.

If $y = 0$, loss becomes $-\ln(1-p)$.

Two predictions, same $p = 0.8$

Actual $y = 1 \rightarrow -\ln(0.8) = 0.223$.

Actual $y = 0 \rightarrow -\ln(0.2) = 1.609$.

The prediction is unchanged; the target changes its meaning.

For a dataset, average the per-example losses.

Categorical cross-entropy (CCE)

One-hot target

For the correct class k: $L = -\ln(p_k)$.

Iris example

True class is Versicolor (class 1); predicted probabilities [0.665,0.245,0.090].

Calculate

Loss = $-\ln(0.245) \approx 1.407$, even though the model predicted Setosa.

Categorical cross-entropy: decode the symbols

Single-label multiclass classification

$L = -\Sigma_k\, y_k \ln(p_k)$

k indexes classes.

y_k is 1 for the correct class and 0 for others (one-hot target).

p_k is the model probability for class k.

Only the correct class contributes to the sum.

Air-quality example

Target = Poor → [0, 0, 1].

Prediction = [0.05, 0.15, 0.80].

$L = -\ln(0.80) \approx 0.223$.

If p(Poor) = 0.10, loss ≈ 2.303.

This is an illustrative example, not an empirical result.

Sparse versus categorical

Categorical cross-entropy

Targets are one-hot vectors such as [0,1,0].

Sparse categorical cross-entropy

Targets are integer class IDs such as 1.

Same concept

For equivalent targets and predictions, they represent the same multiclass log-loss.

Logits or probabilities?

Logits

Unnormalized scores; can be any real number.

Probabilities

Sigmoid or softmax transforms scores into interpretable ranges.

Numerical stability

In Keras, use from_logits=True with raw logits; do not apply softmax twice.

Match the task and output

Regression

Linear output + MSE or MAE are common starting choices.

Binary classification

One sigmoid output + binary cross-entropy.

Exclusive multiclass classification

K class logits + sparse/categorical cross-entropy, or softmax probabilities with matching loss settings.

Compare loss functions by task

The target type determines which errors are meaningful

Regression

MSE: squared numerical error; high outlier sensitivity.

MAE: absolute numerical error; more outlier-resistant.

Huber: quadratic near zero, linear for large residuals; requires a transition parameter δ.

Classification

BCE: binary label with one probability; also used independently per label in multilabel tasks.

CCE: one-hot single-label multiclass target.

Sparse CCE: same multiclass objective, but integer-coded target.

Loss is not accuracy, F1 or calibration.

Why this matters in real systems

Spam filtering

Binary classification: sigmoid and BCE; consider false positives.

Visual recognition

Multiclass image classification: softmax probabilities and cross-entropy.

Sensor forecasting

Predict temperature or energy demand: numerical output and regression loss.

Which loss should we avoid?

The wrong objective can misrepresent the problem

Single-label vs multilabel

Single-label: one category per sample → softmax + CCE (or sparse CCE).

Multilabel: several categories may be true simultaneously → independent sigmoid outputs + BCE.

Softmax would force probabilities to compete even when multiple labels apply.

Regression vs classification

Predict PM2.5 concentration (µg/m³) → numerical regression; consider MSE, MAE or Huber.

Predict one of three predefined air-quality categories → multiclass classification; use CCE.

Category definitions and data labels must be documented.

Where activation functions appear today

Transformer feed-forward blocks

Nonlinear activations such as GELU and gated variants help transform learned features.

Language model training

Next-token prediction commonly uses cross-entropy over token logits.

Important caveat

Large language models also involve attention, normalization and many other components; activation alone does not explain their behavior.

INTERACTIVE

Choose the correct pair

Case A

Predict house energy consumption in kWh. Which output and loss?

Case B

Detect whether a network event is malicious. Which output and loss?

Case C

Classify an Iris flower into one of three species. Which output and loss?

Discussion answers

Case A

Numerical output + MSE or MAE.

Case B

Sigmoid + BCE; tune the decision threshold using validation data.

Case C

Three logits + sparse categorical cross-entropy, or softmax + matching categorical loss.

Calculate the loss by hand

Question 1

y=1 and p=0.8: calculate BCE.

Question 2

y=0 and p=0.8: calculate BCE.

Question 3

Why is the second penalty larger?

Worked answers

Question 1

$-\ln(0.8) \approx 0.223$.

Question 2

$-\ln(0.2) \approx 1.609$.

Reason

The same probability 0.8 is appropriate for y=1 but a confident error for y=0.

A new ANN problem: air quality classification

We use sensor data to classify air quality into different categories. For simplicity, these categories are defined for this exercise rather than based on official AQI (Air Quality Index) thresholds.

Input features

PM2.5 and PM10: particulate concentrations (µg/m³).
Temperature: ambient air temperature (°C).
Relative humidity: percentage (%).
Measurements need timestamps and quality checks.

Target classes

Good / Moderate / Poor: three mutually exclusive target labels.
Labels must be defined from a documented rule or validated observations.
Avoid using input variables to create circular or misleading labels.

Class question: Which inputs could influence air quality, and which variables might be missing?

From sensor readings to class probabilities

One measurement produces four numerical inputs. Standardization may improve training.

INPUT

4 sensor features
PM2.5 · PM10
Temperature · Humidity

→

HIDDEN

Dense(8, ReLU)
Learns nonlinear feature combinations.
8 is an example, not an optimum.

→

OUTPUT

Dense(3, Softmax)
Outputs three probabilities whose sum is 1.

Why ReLU? It introduces nonlinearity. Why Softmax? It gives a probability distribution across the three classes.

Read the model prediction

Illustrative probabilities for one observation; not measurements or a trained-model result.



Prediction: Poor (highest probability). But how do we measure the quality of this prediction?

Cross-entropy: confident versus uncertain

Assume the true class is Poor. Compare the probability assigned to that correct class.

Prediction A: Poor = 0.80

$L = -\ln(0.80) \approx 0.223$

Low loss: the correct class receives high probability.

Prediction B: Poor = 0.10

$L = -\ln(0.10) \approx 2.303$

High loss: the correct class receives low probability.

Discussion: Both models can output a class label. Why does cross-entropy provide more information than accuracy alone?

APPLICATION • CLASS ACTIVITY

Choose the activation and loss

Choose an output layer and training loss for each problem. Explain why the other choice is unsuitable.

Scenario A • Classification

Predict Good / Moderate / Poor from four inputs.
Three exclusive labels.
Which output activation?
Which cross-entropy variant for integer labels?

Scenario B • Regression

Predict a continuous PM2.5 concentration ($\mu g/m^3$).
One numeric target.
Which output activation?
Which regression loss?

Suggested: A → 3 softmax outputs + sparse categorical cross-entropy; B → 1 linear output + MSE or MAE.

Mini exercise: select and explain the loss

Ask students to justify the output and the training objective

Problem A: sensor forecasting

Predict tomorrow’s temperature in °C.

What is the target type?

Would MSE or MAE be preferable if occasional sensor spikes occur?

Discuss how an outlier changes each loss.

Problem B: air-quality labels

Predict Good / Moderate / Poor (one class only).

Choose output activation and loss.

Then change the task: predict multiple simultaneous pollution alerts.

Why should the output/loss pair change?

Extend our previous Iris demo

Experiment 1

Keep the architecture fixed; inspect softmax probabilities and the true labels.

Experiment 2

Record training loss and validation loss for several epochs.

Experiment 3

Compare ReLU and tanh with the same split, seed and training setup; do not assume a winner.

What to look for in loss curves

Both curves decrease

Training is improving; still inspect held-out performance.

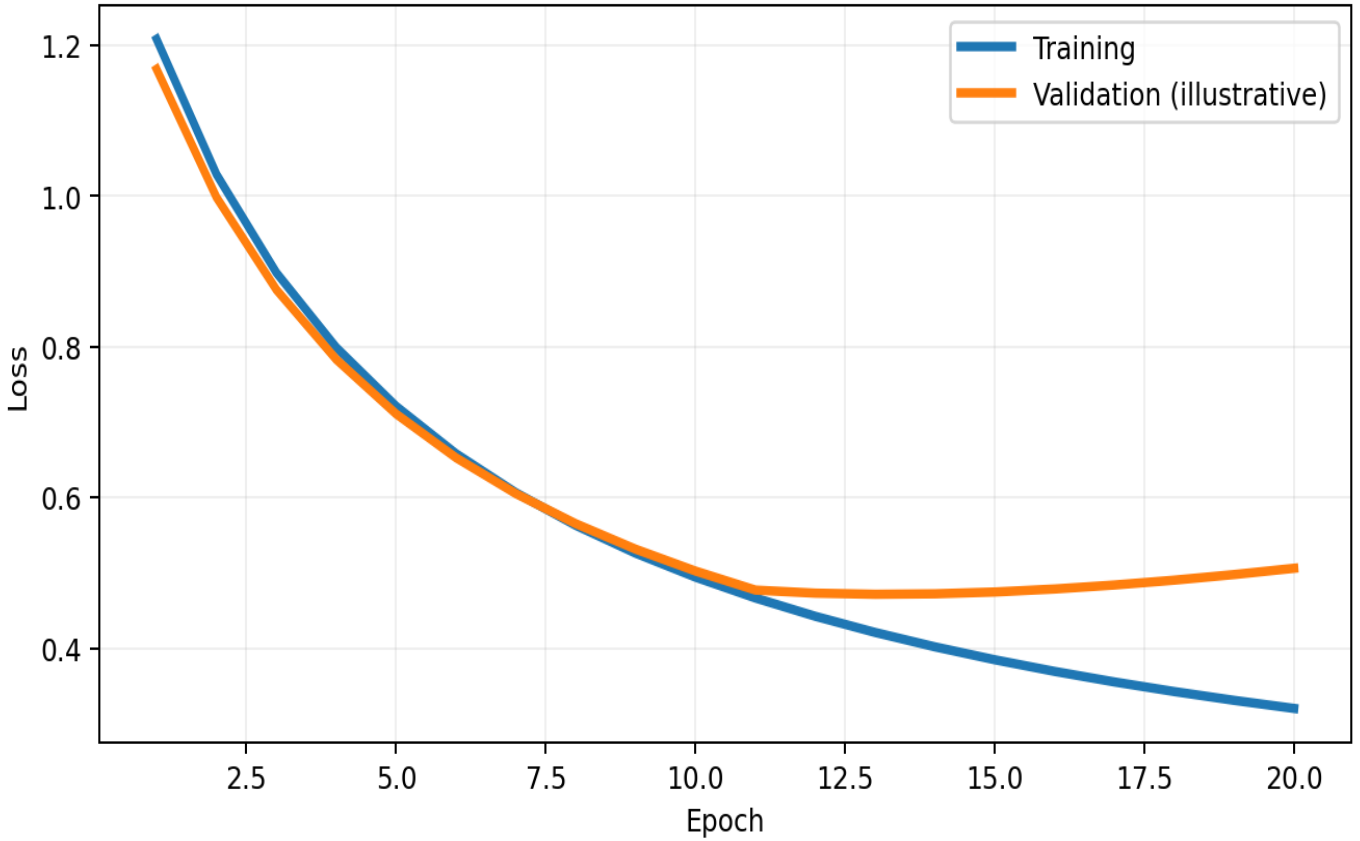
Training decreases; validation rises

Possible overfitting; investigate rather than diagnosing from one noisy epoch.

Both remain high

Possible underfitting, poor settings, or data/label problems.

Reading training and validation loss



Illustrative curves(sample for the lecture), not experimental results

- Training loss keeps decreasing.
- Validation loss eventually rises.
- Potential overfitting: compare held-out data.

EXIT TICKET

Explain in one minute

1

Why do we use a nonlinear activation in hidden layers?

2

What is the difference between Softmax and cross-entropy?

3

Which loss would you choose for a continuous sensor measurement?

Bridge to backpropagation

Forward pass

Produce activations and a prediction.

Loss

Quantify the discrepancy between prediction and target.

Backward pass

Compute how changes to weights affect the loss using the chain rule.

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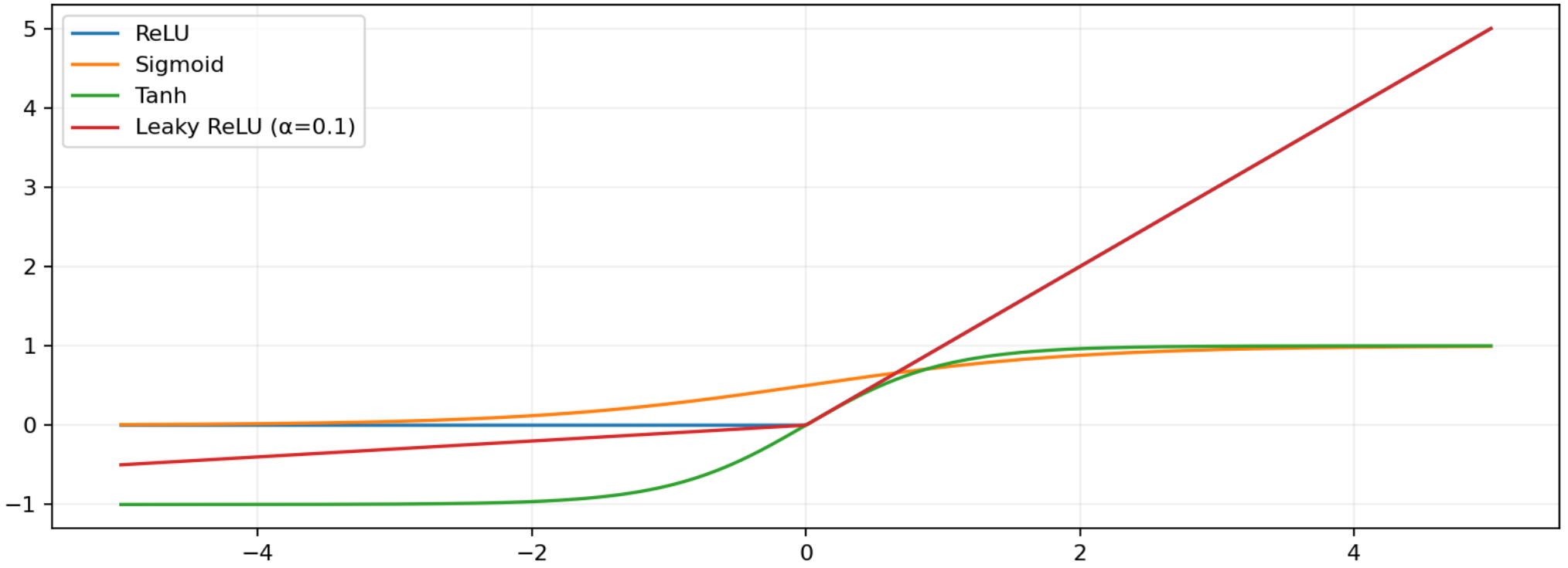
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Activation curves



Loss behavior

